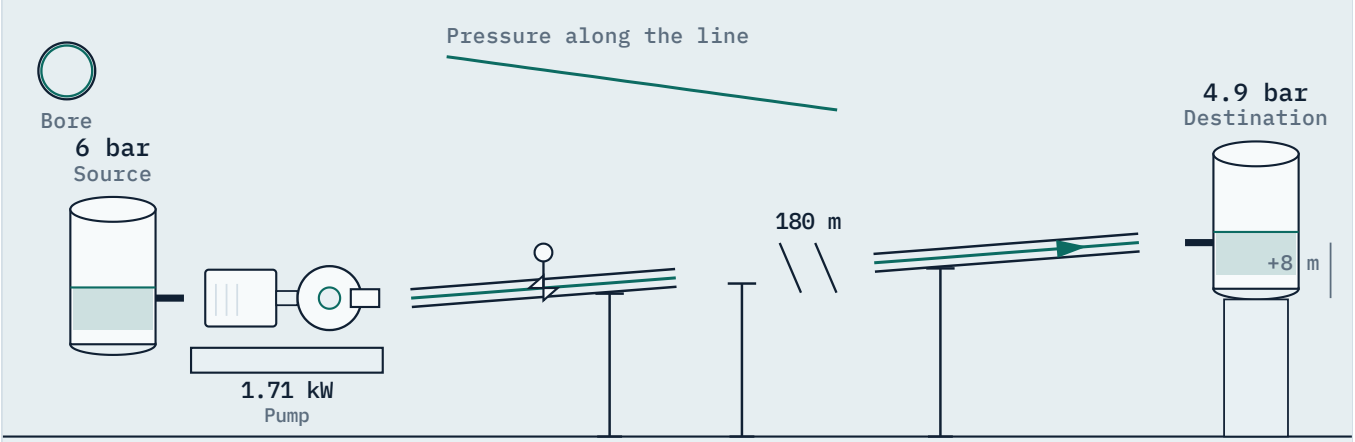
Water line

Liquid line · Water 25°C

PRESSURE DROP	OUTLET	VELOCITY	PUMP
1.1 bar	4.9 bar	1.35 m/s	1.71 kW



Source 6 bar → Destination 4.9 bar

NPS 4 · Sch 40 / STD · ID 102.3 mm · 40 m³/h · 180 m · +8 m · 1.71 kW

How this design sits. Vertical scale is stretched so the lift is visible. The break in a long run means the drawing is not to length.

DESIGN DATA

Flow	40 m³/h
Mass flow	39,880 kg/h
Inside diameter	102.3 mm
Pipe	NPS 4 · Sch 40 / STD
Length	180 m (180 m)
Elevation	+8 m (8 m)
Inlet, gauge	6 bar
Density	997 kg/m³
Viscosity	0.89 cP
Roughness	Commercial steel · 0.046 mm
Fittings	2 × 90° elbow LR, 1 × Gate valve · ΣK 1.07
Velocity band	Process liquid 1–3 m/s
Pump / motor	75% / 95%
Duty	8,000 h/y · 100% · 0.29 AED/kWh

CALCULATION

1	Area π D² / 4	8.21e-3 m²
2	Velocity Q / A	1.35 m/s
3	Reynolds ρ V D / μ	1.55e+5
4	Darcy f, Churchill · turbulent	0.0192
5	Friction f (L/D) ρ V² / 2	0.31 bar
6	Fittings ΣK ρ V² / 2 · ΣK 1.07	0.01 bar
7	Elevation ρ g Δz	0.78 bar
8	Total drop friction + fittings + lift	1.1 bar
9	Outlet inlet − total drop	4.9 bar
10	Head ΔP / (ρ g)	11.25 m
11	Pump ρ g Q H / (ηp ηm) · 75% × 95%	1.71 kW

OPTIMUM BORE AND ENERGY

This bore 102.3 mm · 1.71 kW · 13,233.3 AED/y (9,255 pipe + 3,978.3 power)

Economic bore 91.15 mm · 2.1 kW · 12,885 AED/y (8,015.8 pipe + 4,869.2 power)

Towler & Sinnott carbon-steel screen 188.5 mm. Ignores your tariff. Not for gas, slurry or alloy pipe.

FLOW VARIANTS · 60% TO 140%

Flow	ΔP	Outlet	m/s
24	**0.9**	**5.1**	**0.81**
32	**0.99**	**5.01**	**1.08**
40	**1.1**	**4.9**	**1.35**
48	**1.23**	**4.77**	**1.62**
56	**1.38**	**4.62**	**1.89**

Darcy–Weisbach. Friction factor is Churchill (1977). Pump kilowatts are electric power for the whole line, including elevation and fittings.

The sketch is how the line sits: equipment at the ends, pipe on supports, lift in the grade or the rack. Vertical scale is stretched. Length is not to that scale.

Library fluid properties. The tariff is the one entered on the cost basis. Screening study — not issued for construction.

Prepared

Checked

Date